

OSSP PRACTICAL 7

```
alurusaranya@LAPTOP-0PA61QEN:~$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ ^C
alurusaranya@LAPTOP-0PA61QEN:~$ mkdir practical7
alurusaranya@LAPTOP-0PA61QEN:~$ cd practical7
alurusaranya@LAPTOP-0PA61QEN:~/practical7$ nano memory.c
alurusaranya@LAPTOP-0PA61QEN:~/practical7$ |
```

```
alurusaranya@LAPTOP-0PA61QEN:~/practical7$ ./memory
Address of code: 0x6171d4832209
Address of global: 0x6171d4835010
Address of static: 0x6171d4835014
Address of heap: 0x6171f4c3d2a0
Address of stack: 0x7fff96f4fd1c
Process ID: 754
Press Enter to exit...
cat /proc/754/maps
alurusaranya@LAPTOP-0PA61QEN:~/practical7$ pamp 754
Command 'pamp' not found, but can be installed with:
sudo apt install paml
alurusaranya@LAPTOP-0PA61QEN:~/practical7$ pmap 754
alurusaranya@LAPTOP-0PA61QEN:~/practical7$
```

```
alurusaranya@LAPTOP-0PA61QEN:~/practical7$ pmap 754
alurusaranya@LAPTOP-0PA61QEN:~/practical7$ cd ~/practical7
alurusaranya@LAPTOP-0PA61QEN:~/practical7$ ./memory
Address of code: 0x629cfec55209
Address of global: 0x629cfec58010
Address of static: 0x629cfec58014
Address of heap: 0x629d2f0242a0
Address of stack: 0x7ffc44ea885c
Process ID: 876
Press Enter to exit...
alurusaranya@LAPTOP-0PA61QEN:~/practical7$
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;

void code_function()
{
    printf("Address of code:  %p\n", (void *)code_function);
}

int main()
{
    static int static_var = 20;
    int stack_var = 30;
    int *heap_var = malloc(sizeof(int));

    *heap_var = 40;

    code_function();

    printf("Address of global:  %p\n", (void *)&global_var);
    printf("Address of static:  %p\n", (void *)&static_var);
    printf("Address of heap:    %p\n", (void *)heap_var);
    printf("Address of stack:   %p\n", (void *)&stack_var);

    printf("Process ID: %d\n", getpid());

    printf("Press Enter to exit...\n");
    getchar();

    free(heap_var);
}
```

```

int main()
{
    static int static_var = 20;
    int stack_var = 30;
    int *heap_var = malloc(sizeof(int));

    *heap_var = 40;

    code_function();

    printf("Address of global: %p\n", (void *)&global_var);
    printf("Address of static: %p\n", (void *)&static_var);
    printf("Address of heap: %p\n", (void *)heap_var);
    printf("Address of stack: %p\n", (void *)&stack_var);

    printf("Process ID: %d\n", getpid());

    printf("Press Enter to exit...\n");
    getchar();

    free(heap_var);

    return 0;
}

```

TERMINAL 2

```

root@LAPTOP-0PA61QEN:~# ls /home
alurusaranya
root@LAPTOP-0PA61QEN:~# su alurusaranya
alurusaranya@LAPTOP-0PA61QEN:/root$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ pmap 876
876:  /memory
0000629cfec54000      4K r---- memory
0000629cfec55000      4K r-x-- memory
0000629cfec56000      4K r---- memory
0000629cfec57000      4K r---- memory
0000629cfec58000      4K rw--- memory
0000629d2f024000    132K rw--- [ anon ]
00007c875ea00000    160K r---- libc.so.6
00007c875ea28000   1568K r-x-- libc.so.6
00007c875ebb0000    316K r---- libc.so.6
00007c875ebff000     16K r---- libc.so.6
00007c875ec03000      8K rw--- libc.so.6
00007c875ec05000     52K rw--- [ anon ]
00007c875edb4000     12K rw--- [ anon ]
00007c875edbc000      8K rw--- [ anon ]
00007c875edbe000     16K r---- [ anon ]
00007c875edc2000      8K r---- [ anon ]
00007c875edc4000      8K r-x-- [ anon ]
00007c875edc6000      4K r---- ld-linux-x86-64.so.2
00007c875edc7000    172K r-x-- ld-linux-x86-64.so.2
00007c875edf2000     40K r---- ld-linux-x86-64.so.2
00007c875edfc000      8K r---- ld-linux-x86-64.so.2
00007c875edfe000      8K rw--- ld-linux-x86-64.so.2
00007ffc44e89000    132K rw--- [ stack ]
total                2688K
alurusaranya@LAPTOP-0PA61QEN:~$ cat /proc/876/maps
629cfec54000-629cfec55000 r-p 00000000 08:30 14553 /home/alurusaranya/practical7/memory
629cfec55000-629cfec56000 r-xp 00001000 08:30 14553 /home/alurusaranya/practical7/memory
629cfec56000-629cfec57000 r-p 00002000 08:30 14553 /home/alurusaranya/practical7/memory
629cfec57000-629cfec58000 r-p 00002000 08:30 14553 /home/alurusaranya/practical7/memory
629cfec58000-629cfec59000 rw-p 00003000 08:30 14553 /home/alurusaranya/practical7/memory
629d2f024000-629d2f045000 rw-p 00000000 00:00 0 [heap]
7c875ea00000-7c875ea28000 r-p 00000000 08:30 49795 /usr/lib/x86_64-linux-gnu/libc.so.6
7c875ea28000-7c875ebb0000 r-xp 00028000 08:30 49795 /usr/lib/x86_64-linux-gnu/libc.so.6
7c875ebb0000-7c875ebff000 r-p 001b0000 08:30 49795 /usr/lib/x86_64-linux-gnu/libc.so.6
7c875ebff000-7c875ec03000 r-p 001fe000 08:30 49795 /usr/lib/x86_64-linux-gnu/libc.so.6
7c875ec03000-7c875ec05000 rw-p 00202000 08:30 49795 /usr/lib/x86_64-linux-gnu/libc.so.6
7c875ec05000-7c875ec12000 rw-p 00000000 00:00 0
7c875edb4000-7c875edb7000 rw-p 00000000 00:00 0
7c875edbc000-7c875edbe000 rw-p 00000000 00:00 0

```

```

alurusaranya@LAPTOP-0PA61QEN:~$ cat /proc/876/maps
629cfec54000-629cfec55000 r-p 00000000 08:30 14553 /home/alurusaranya/practical7/memory
629cfec55000-629cfec56000 r-xp 00001000 08:30 14553 /home/alurusaranya/practical7/memory
629cfec56000-629cfec57000 r-p 00002000 08:30 14553 /home/alurusaranya/practical7/memory
629cfec57000-629cfec58000 r-p 00002000 08:30 14553 /home/alurusaranya/practical7/memory
629cfec58000-629cfec59000 rw-p 00003000 08:30 14553 /home/alurusaranya/practical7/memory
629d2f024000-629d2f045000 rw-p 00000000 00:00 0 [heap]
7c875ea00000-7c875ea28000 r-p 00000000 08:30 49795 /usr/lib/x86_64-linux-gnu/libc.so.6
7c875ea28000-7c875ebb0000 r-xp 00028000 08:30 49795 /usr/lib/x86_64-linux-gnu/libc.so.6
7c875ebb0000-7c875ebff000 r-p 001b0000 08:30 49795 /usr/lib/x86_64-linux-gnu/libc.so.6
7c875ebff000-7c875ec03000 r-p 001fe000 08:30 49795 /usr/lib/x86_64-linux-gnu/libc.so.6
7c875ec03000-7c875ec05000 rw-p 00202000 08:30 49795 /usr/lib/x86_64-linux-gnu/libc.so.6
7c875ec05000-7c875ec12000 rw-p 00000000 00:00 0
7c875edb4000-7c875edb7000 rw-p 00000000 00:00 0
7c875edbc000-7c875edbe000 rw-p 00000000 00:00 0
7c875edbe000-7c875edc2000 r-p 00000000 00:00 0 [vvar]
7c875edc2000-7c875edc4000 r-p 00000000 00:00 0 [vvar_vclock]
7c875edc4000-7c875edc6000 r-xp 00000000 00:00 0 [vdso]
7c875edc6000-7c875edc7000 r-p 00000000 08:30 49792 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7c875edc7000-7c875edf2000 r-xp 00001000 08:30 49792 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7c875edf2000-7c875edfc000 r-p 0002c000 08:30 49792 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7c875edfc000-7c875edfe000 r-p 00036000 08:30 49792 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7c875edfe000-7c875ee00000 rw-p 00038000 08:30 49792 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffc44e89000-7ffc44eaa000 rw-p 00000000 00:00 0 [stack]

```

```

alurusaranya@LAPTOP-0PA61QEN:~$ cat /proc/876/status
Name:  memory
Umask: 0002
State: S (sleeping)
Tgid: 876
Ngid: 0
Pid: 876
PPid: 725
TracerPid: 0
Uid: 1000 1000 1000 1000
Gid: 1000 1000 1000 1000
FDSize: 256
Groups: 4 20 24 25 27 29 30 44 46 100 107 1000
NSTgid: 876
NSpid: 876
NSpgid: 876
NSsid: 704
Kthreid: 0

```

[illegible]

```
alurusarany@LAPTOP-8PK61QEN:~$ cat /proc/876/smaps
629cfe54000-629cfe55000 r--p 00000000 08:30 14553
Size: 4 kB
KernelPageSize: 4 kB
MMUPageSize: 4 kB
Rss: 4 kB
Pss: 4 kB
Pss_Dirty: 0 kB
Shared_Clean: 0 kB
Shared_Dirty: 0 kB
Private_Clean: 4 kB
Private_Dirty: 0 kB
Referenced: 4 kB
Anonymous: 0 kB
KSM: 0 kB
LazyFree: 0 kB
AnonHugePages: 0 kB
ShmemPmdMapped: 0 kB
FilePmdMapped: 0 kB
Shared_Hugetlb: 0 kB
Private_Hugetlb: 0 kB
Swap: 0 kB
SwapPss: 0 kB
Locked: 0 kB
THPeligible: 0
VmFlags: rd mr mw me sd
629cfe55000-629cfe56000 r-xp 00001000 08:30 14553
Size: 4 kB
KernelPageSize: 4 kB
MMUPageSize: 4 kB
Rss: 4 kB
Pss: 4 kB
Pss_Dirty: 0 kB
Shared_Clean: 0 kB
Shared_Dirty: 0 kB
Private_Clean: 4 kB
Private_Dirty: 0 kB
Referenced: 4 kB
Anonymous: 0 kB
KSM: 0 kB
LazyFree: 0 kB
AnonHugePages: 0 kB
ShmemPmdMapped: 0 kB
FilePmdMapped: 0 kB
```

```

629cfec56000-629cfec57000 r--p 00002000 08:30 14553 /home/alurusaranya/practical7/memory
Size: 4 kB
KernelPageSize: 4 kB
MMUPageSize: 4 kB
Rss: 4 kB
Pss: 4 kB
Pss_Dirty: 0 kB
Shared_Clean: 0 kB
Shared_Dirty: 0 kB
Private_Clean: 4 kB
Private_Dirty: 0 kB
Referenced: 4 kB
Anonymous: 0 kB
KSM: 0 kB
LazyFree: 0 kB
AnonHugePages: 0 kB
ShmemPmdMapped: 0 kB
FilePmdMapped: 0 kB
Shared_Hugetlb: 0 kB
Private_Hugetlb: 0 kB
Swap: 0 kB
SwapPss: 0 kB
Locked: 0 kB
THPeligible: 0
VmFlags: rd mr mm me sd
629cfec57000-629cfec58000 r--p 00002000 08:30 14553 /home/alurusaranya/practical7/memory
Size: 4 kB
KernelPageSize: 4 kB
MMUPageSize: 4 kB
Rss: 4 kB
Pss: 4 kB
Pss_Dirty: 4 kB
Shared_Clean: 0 kB
Shared_Dirty: 0 kB
Private_Clean: 0 kB
Private_Dirty: 4 kB
Referenced: 4 kB
Anonymous: 4 kB
KSM: 0 kB
LazyFree: 0 kB
AnonHugePages: 0 kB
ShmemPmdMapped: 0 kB
FilePmdMapped: 0 kB
Shared_Hugetlb: 0 kB
Private_Hugetlb: 0 kB

```

```

alurusaranya@LAPTOP-0PA61QEN:~$ cat /proc/876/status | grep -E "VmSize|VmRSS|VmData|VmStk|VmExe"
VmSize: 2688 kB
VmRSS: 1544 kB
VmData: 224 kB
VmStk: 132 kB
VmExe: 4 kB
alurusaranya@LAPTOP-0PA61QEN:~$ ps -p 876 -o pid,ppid,state,%mem,rss,vsz,cmd
PID PPID S %MEM RSS VSZ CMD
876 725 S 0.0 1544 2688 ./memory
alurusaranya@LAPTOP-0PA61QEN:~$ id
uid=1000(alurusaranya) gid=1000(alurusaranya) groups=1000(alurusaranya),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),107(netdev)
alurusaranya@LAPTOP-0PA61QEN:~$ cat /proc/876/stat
876 (memory) S 725 876 704 34816 876 4194304 84 0 0 0 0 0 0 0 0 20 0 1 0 128005 2752512 346 18446744073709551615 108426428764160 108426428765005 140721464707472 0 0 0 0 0 1 0 17 5 0 0 0 0 0 108426428775816 108426428776472 108427238064128 140721464710512 140721464710521 140721464710521 140721464713199 0
alurusaranya@LAPTOP-0PA61QEN:~$ readlink /proc/876/exe
/home/alurusaranya/practical7/memory
alurusaranya@LAPTOP-0PA61QEN:~$ readlink /proc/876/exe
/home/alurusaranya/practical7/memory
alurusaranya@LAPTOP-0PA61QEN:~$ /home/alurusaranya/practical7/memory
Address of code: 0x5bb5d3855209
Address of global: 0x5bb5d3858010
Address of static: 0x5bb5d3858014
Address of heap: 0x5bb5f70b92a0
Address of stack: 0x7ffff301f7bc
Process ID: 932
Press Enter to exit...
alurusaranya@LAPTOP-0PA61QEN:~$ readlink /proc/876/cwd
/home/alurusaranya/practical7
alurusaranya@LAPTOP-0PA61QEN:~$ cat /proc/876/cmdline
./memoryalurusaranya@LAPTOP-0PA61QEN:~$ ps -p 876 -o pid,%mem,vsz,rss,cmd
PID %MEM VSZ RSS CMD
876 0.0 2688 1544 ./memory
alurusaranya@LAPTOP-0PA61QEN:~$ ps -p 876
PID TTY TIME CMD
alurusaranya@LAPTOP-0PA61QEN:~$ pmap

Usage:
pmap [options] PID [PID ...]

```

```

alurusaranya@LAPTOP-0PA61QEN:~$ readlink /proc/876/cwd
/home/alurusaranya/practical7
alurusaranya@LAPTOP-0PA61QEN:~$ cat /proc/876/cmdline
./memory
alurusaranya@LAPTOP-0PA61QEN:~$ ps -p 876 -o pid,%mem,vsize,ss,cmd
  PID %MEM    VSZ   RSS CMD
   876   0.0  2688  1544 ./memory
alurusaranya@LAPTOP-0PA61QEN:~$ ps -p 876
  PID TTY          TIME CMD
   876   -        0:00 ./memory
alurusaranya@LAPTOP-0PA61QEN:~$ pmap

Usage:
pmap [options] PID [PID ...]

Options:
-x, --extended          show details
-X                      show even more details
-XX                     WARNING: format changes according to /proc/PID/smaps
                        show everything the kernel provides
-c, --read-rc           read the default rc
-C, --read-rc-from=<file> read the rc from file
-n, --create-rc         create new default rc
-N, --create-rc-to=<file> create new rc to file
                        NOTE: pid arguments are not allowed with -n, -N
-d, --device           show the device format
-q, --quiet            do not display header and footer
-p, --show-path         show path in the mapping
-A, --range=<low>[,<high>] limit results to the given range

-h, --help            display this help and exit
-V, --version         output version information and exit

For more details see pmap(1).
alurusaranya@LAPTOP-0PA61QEN:~$ |

```

Practical 7 – Theory: Process Memory and Address Space

1. Process Memory

A **process** is a program that is currently executing. Each process gets its own **virtual address space**, which contains different regions used for code, data, dynamic memory, and function calls.

2. Main Memory Regions

A process's address space is generally divided into:

- **Text/Code segment** – contains the machine instructions of the program.
- **Data segment** – contains global and static variables.
- **Heap** – used for **dynamic memory allocation**, such as memory allocated using `malloc()`.
- **Stack** – contains local variables, function parameters, and function-call information.

3. Global Variables

A global variable is declared outside functions.

Example:

```
int global_var = 10;
```

It normally resides in the **data segment**.

4. Static Variables

A static variable has a lifetime that lasts for the entire execution of the program.

Example:

```
static int static_var = 20;
```

It is stored in the **data area** of the process.

5. Stack Memory

Local variables declared inside functions normally use stack memory.

Example:

```
int stack_var = 30;
```

The stack is also used when functions are called and returned.

6. Heap Memory

The heap is used when memory is allocated dynamically.

Example:

```
int *p = malloc(sizeof(int));
```

The memory obtained using malloc() comes from the **heap** and should be released using:

```
free(p);
```

7. /proc Filesystem

Linux provides information about running processes through the **/proc filesystem**.

For a process with PID 876:

```
cat /proc/876/maps
```

shows the process's memory mappings.

```
cat /proc/876/status
```

shows information such as process state and memory usage.

8. pmap

The pmap command displays the memory map of a process.

```
pmap 876
```

It helps identify regions such as:

- program code
- heap
- stack
- shared libraries

9. Memory Permissions

In memory maps, permissions can include:

- r → Read
- w → Write
- x → Execute
- p → Private mapping

For example:

r-xp

means the region is **readable and executable** and privately mapped.

10. Process ID (PID)

Every running process has a unique **Process ID (PID)**.

In C:

getpid()

returns the PID of the current process.

11. Purpose of the Practical

The practical demonstrates how to:

1. Create a C program containing global, static, stack, and heap variables.
2. Display their memory addresses.
3. Obtain the process PID.
4. Examine the process's memory layout using /proc.
5. Analyze memory regions using pmap.